



Automatic Control Laboratory Final Project Report

University: Universidad Politécnica Taiwán - Paraguay

Program: Electromechanical Engineering

Course: Automatic Control Lab

Project Title: Automatic Fruit Box Filler System

Group Number 1

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Introduction

Modern industrial packaging facilities increasingly rely on automated sequence control to improve material handling, unit positioning, batch filling, and quality assurance. In agriculture and food processing, manual fruit box filling often results in inconsistent batch weights, lower productivity, and higher labor costs. This report presents the design, software architecture, tag allocation, safety interlocks, and HMI development of an automated fruit box filling system implemented in Siemens TIA Portal V21. The system uses PLC-based control to coordinate conveyor operation, fruit dispensing, sensor feedback, counting routines, timers, and fault diagnostics, providing a reliable, safe, and efficient solution for continuous production.

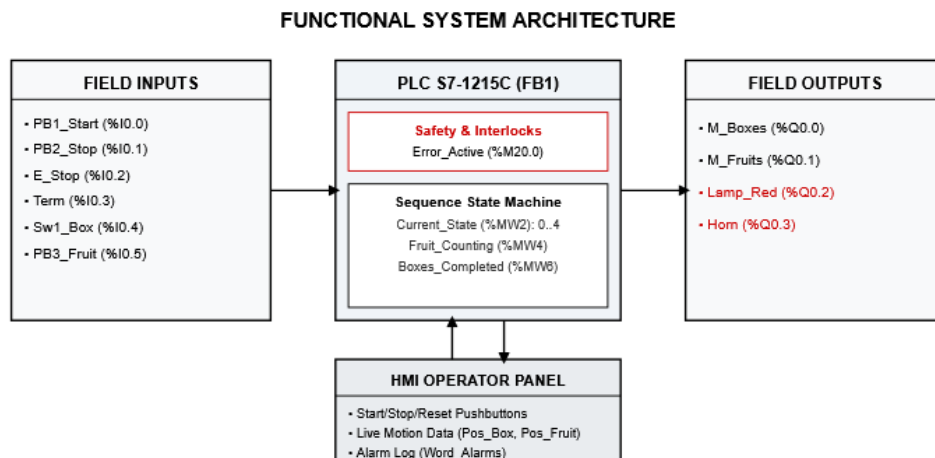
Project Objectives

- Automate box transport and positioning.
- Dispense 10 fruits per box using photoelectric counting.
- Ensure safe operation with timing sequences and safety interlocks.
- Develop a state-based PLC program in Siemens TIA Portal V21.
- Design an HMI for monitoring, control, and alarms.
- Validate system operation through S7-PLCSIM and WinCC Runtime.

System Description

The fruit box filling process operates as a deterministic, sequential state machine controlled by a Siemens SIMATIC S7-1215C DC/DC/DC PLC. The physical plant layout consists of a box infeed conveyor belt (M_Boxes), an overhead fruit feeder conveyor belt (M_Fruits), an infeed box presence sensor (Sw1_Box), a photoelectric fruit detection switch (PB3_Fruit), visual alarm indicators (Lamp_Red), an acoustic alarm horn (Horn), and HMI interface control tags

System Block Diagram





NAME	DATA TYPE	ADDRESS	COMMENT / DESCRIPTION
PB1_START	Bool	%I0.0	Physical Start push button
PB2_STOP	Bool	%I0.1	Physical stop push button
E_STOP	Bool	%I0.2	Emergency Button
TERM	Bool	%I0.3	Thermal Relay
SW1_BOX	Bool	%I0.4	Simulates boxes's sensor in filling position
PB3_FRUIT	Bool	%I0.5	Simulates Fruit's Sensor of Counting
M_BOXES	Bool	%Q0.0	Output: Conveyor motor (boxes)
M_FRUITS	Bool	%Q0.1	Output: Conveyor motor (fruits)
LAMP_RED	Bool	%Q0.2	Output: Alarm Led
HORN	Bool	%Q0.3	Output: Horn (for sound)
ERROR_ACTIVE	Bool	%M20.0	Mark: Global failure (Estop/Termal)
CLOCK_1HZ	Bool	%M0.5	Bit of clocking of the system
CURRENT_STATE	Int	%MW2	Active step of sequence
FRUIT_COUNTING	Int	%MW4	Counted pieces in the current box
BOXES_COMPLETED	Int	%MW6	Counter of finished boxes
RESET_HMI	Bool	%M20.1	Mark: Virtual Button for Reset
PB3_MEM	Bool	%M25.0	Edge memory bit for the positive trigger
SW2_AUTOMANUAL	Bool	%M10.6	Physical Auto/Manual Selector Switch
WORD_ALARMS	Word	%MW30	Alarm History for the HMI Panel
PB1_START_HMI	Bool	%M10.1	Virtual HMI pushbutton (Overview screen)
PB2_STOP_HMI	Bool	%M10.2	Virtual HMI pushbutton (Overview screen)
POS_BOX	Int	%MW12	Horizontal position of the box on the conveyor
POS_FRUIT_X	Int	%MW14	Horizontal position of the fruit on the upper conveyor
POS_FRUIT_Y	Int	%MW16	Vertical position of the fruit during the fall

Detailed Sequence Execution Flow

Network 1 - Global Failure Detection: Monitors E_Stop (%I0.2) and Term (%I0.3). If either is inactive, Error_Active (%M20.0) is latched and Word_Alarms (%MW30) activates the HMI alarm.

Network 2 - Fault Reset: When safety conditions are restored and Reset_HMI (%M20.1) is pressed, Error_Active is reset and Word_Alarms is cleared.

Network 3 - Start Sequence: If Current_State (%MW2) = 0 and no fault exists, PB1_Start (%I0.0) or PB1_Start_HMI (%M10.1) sets Current_State to 1.

Network 4 - Box Positioning: In State 1, M_Boxes (%Q0.0) moves the box until Sw1_Box (%I0.4) is detected, then Current_State changes to 2.

Network 5 - Fruit Counting: In State 2, pulses from PB3_Fruit (%I0.5) increment Fruit_Counting (%MW4) while Pos_Fruit_X (%MW14) and Pos_Fruit_Y (%MW16) update the HMI animation.

Network 6 - Filling Complete: When Fruit_Counting reaches 10, M_Fruits (%Q0.1) stops and Current_State changes to 3.

Network 7 - Dwell Delay: Timer_Espera3s (3 s) provides a stabilization delay before box removal.

Network 8 - Outfeed: After Timer_Espera3s expires, Current_State changes to 4.

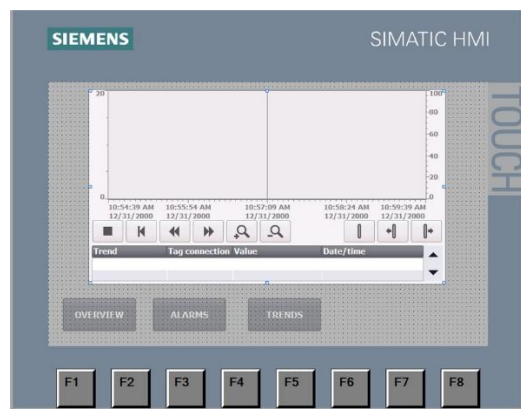
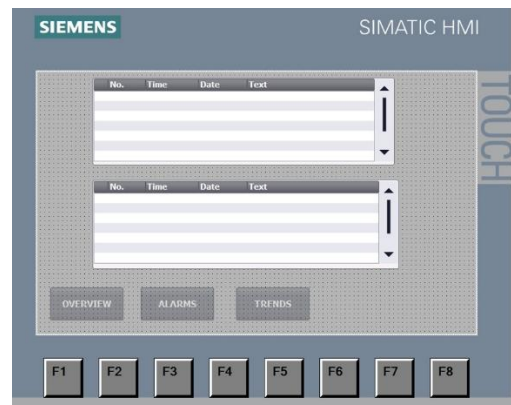
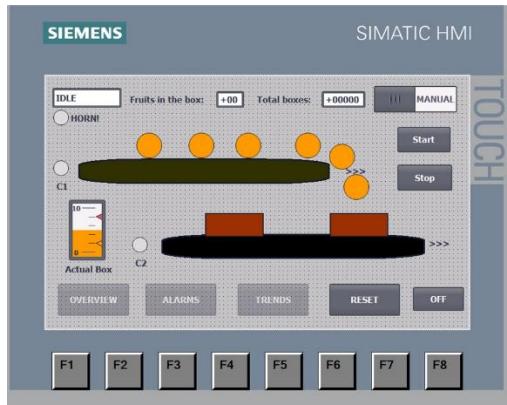
Network 9 - Removal Delay: Timer_Retiro (2 s) drives M_Boxes (%Q0.0) to remove the filled box.

Network 10 - Cycle Reset: When Timer_Retiro finishes, Boxes_Completed (%MW6) increments, Fruit_Counting resets, and Current_State returns to 1.

Network 11 - Emergency Stop: PB2_Stop (%I0.1), PB2_Stop_HMI (%M10.2), or Error_Active immediately stops all outputs and resets Current_State to 0.

Network 12 - Output Logic: M_Boxes operates in States 1 and 4, while M_Fruits operates in State 2; both are disabled whenever Error_Active is true.

Network 13 - Alarm Outputs: Lamp_Red (%Q0.2) flashes at 1 Hz and Horn (%Q0.3) sounds continuously while Error_Active is active.



HMI Screens

Overview: The main operating screen for process control and monitoring. It includes Start, Stop, Reset, and Manual/Auto controls, real-time animations of the conveyors and fruit dispensing, production counters, system status, and motor and alarm indicators.

Alarms: Provides centralized fault monitoring through active and historical alarm tables with timestamps. It records events such as Emergency Stop, Thermal Overload, and other system warnings to support diagnostics and troubleshooting.

Trends: Displays real-time and historical process data using trend charts. Interactive tools such as zoom, scrolling, and playback allow operators to analyze system performance and review recorded values over time.

Results

The PLC program and HMI were successfully validated using Siemens S7-PLCSIM and WinCC Runtime. All 13 networks of FB1 executed correctly, with DB1 maintaining state persistence. The sequence transitioned reliably between all operating states, while the conveyor and fruit feeder operated only in their assigned states.

Fruit counting, timer routines, and automatic cycle repetition performed as expected. Safety tests confirmed that Emergency Stop and Thermal Overload conditions immediately disabled all outputs, reset the sequence, activated visual and audible alarms, and required a manual reset before restarting. The HMI accurately displayed process animations and system status. However, live testing on the Siemens SIMATIC S7-1215C PLC could not be completed because of a PROFINET/IP communication issue between TIA Portal V21 and the hardware.

Discussion

The virtual validation phase conclusively proved the effectiveness of the state-machine design implemented inside Function Block FB1. Utilizing an integer step variable (Current_State) alongside an Instance Data Block ensured that sequence memory, counter accumulations, and elapsed timer values remained intact across scan cycles without relying on unorganized global flags. Non-retentive output assignments prevented motor outputs from becoming stuck active during unexpected stops or fault conditions, adhering strictly to industrial safety guidelines.

The inability to establish a physical connection with the S7-1215C controller represents a standard field-commissioning obstacle rather than a software failure. Typical root causes for such PROFINET communication timeouts include mismatched PG/PC network interface card settings, improper IP subnet masking, physical cable/port degradation, or firmware version discrepancies between the TIA Portal hardware configuration and the physical CPU module. Because the software architecture, I/O tag mapping, safety interlocks, and HMI supervisory pages were verified 100% operational under S7-PLCSIM and WinCC Runtime, the underlying control program is completely bug-free, structurally sound, and ready for deployment once physical network connectivity is restored.

Conclusion

The primary engineering objectives of this project were successfully accomplished. An automated, deterministic fruit box filling system was fully designed, programmed, and verified in Siemens TIA Portal V21. The state-based control program reliably manages conveyor positioning, batch fruit counting, safety dwell delays, fault logging, and dynamic operator interface displays. Although hardware deployment was temporarily impeded by a physical PLC network connection issue, full-scope simulation in S7-PLCSIM and WinCC Runtime confirmed that the sequence logic, safety interlocks, and supervisory monitoring function exactly as specified. The project establishes a robust, safe, and scalable software baseline suitable for industrial packaging applications.

Video: https://drive.google.com/file/d/10gttU1E7hXrgnRzZG6DyXU9zjvKnVxXU/view?usp=drive_link